

Basic Voice

MONOTRAIL TECH TALK

The diagram illustrates the signal flow from the BEATSTEP PRO SEQUENCER 1 to the OSCILLATOR, FILTER, VCA, and ENVELOPE modules. The sequencer has three outputs: PITCH (blue), VELOCITY (blue), and GATE (orange). The OSCILLATOR module has inputs for 1v/oct (blue), FM (black), and Shape (black), and an output labeled 'Out' (pink). The FILTER module has inputs for In (pink), Freq CV (blue), and Res CV (black), and an output labeled 'Out' (pink). The VCA module has inputs for In (pink) and CV (blue), and an output labeled 'Out' (pink). The ENVELOPE module has inputs for Gate (orange), Trig (black), Attack CV (black), and Decay CV (black), and an output labeled 'Out' (pink). The VCA module has an input for In (pink) and an output labeled 'Out' (pink). The signal flow is as follows: PITCH (blue) connects to 1v/oct (blue) of the OSCILLATOR. VELOCITY (blue) connects to Freq CV (blue) of the FILTER and CV (blue) of the VCA. GATE (orange) connects to Gate (orange) of the ENVELOPE. The output of the OSCILLATOR (pink) connects to In (pink) of the FILTER. The output of the FILTER (pink) connects to In (pink) of the VCA. The output of the VCA (pink) connects to In (pink) of the ENVELOPE. The output of the ENVELOPE (pink) connects to In (pink) of the VCA. The output of the VCA (pink) is shown with a downward arrow.

a - This is a traditional synth voice with an oscillator, filter, and amplifier, controlled with the Beatstep Pro. The Beatstep Pro's pitch output is patched to the 1v/oct input of the oscillator. The gate is patched to an envelope modulating the VCA to create a volume curve. And if you want to add velocity to this patch you could add another VCA at end of the audio chain and patch the velocity output of the Beatstep Pro to that VCA, to control the master volume of your patch.

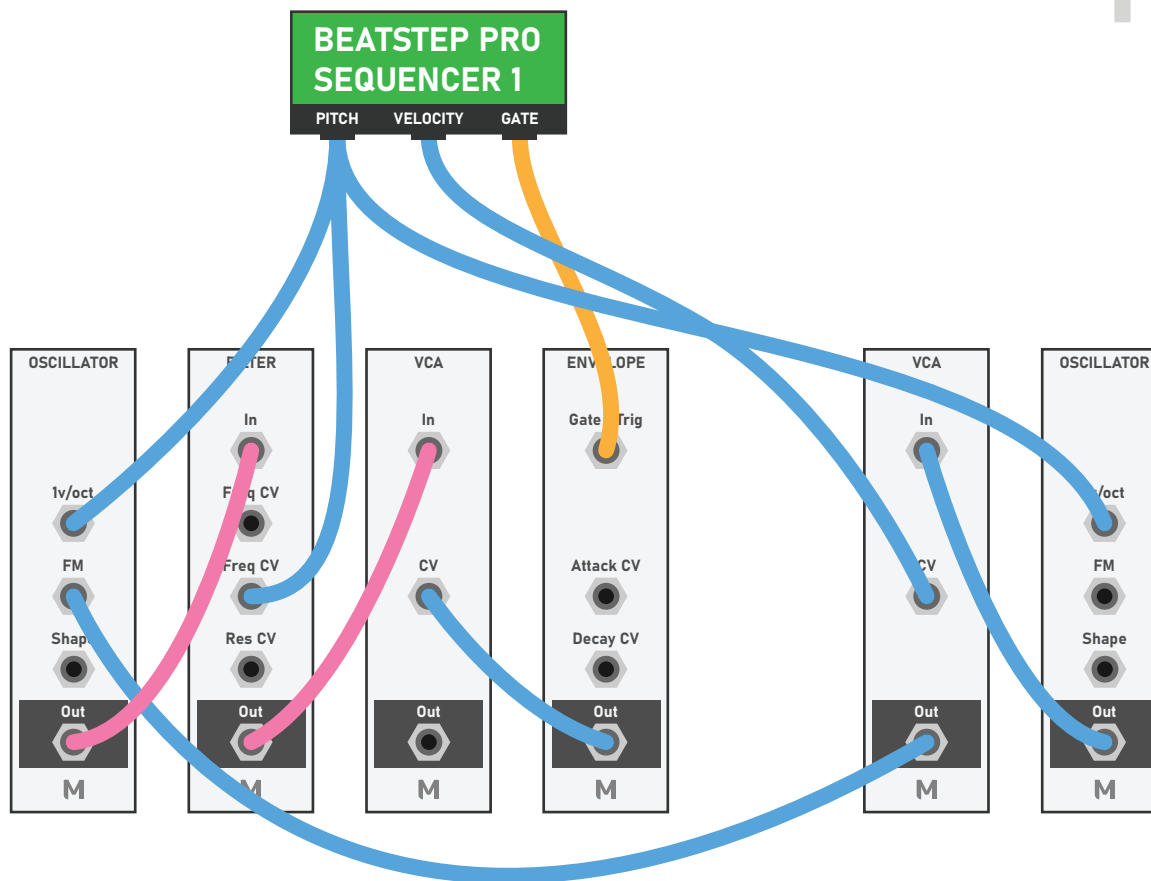
Especially if you use a sequence with a wide range, you can make a copy of the pitch output and send it to a different parameter in the same voice. The filter is a classic destination. This way you open up the filter when playing higher notes, but you can send it anywhere you like.

Eurorack + Beatstep Pro

Creative Velocity

MONOTRAIL TECH TALK

Audio
Control Voltage
Trigger / Gate



(a)

If you don't use the velocity output for volume, you can use this sequence to control a completely different parameter in your voice, like a filter. But to give you a few more ideas to get started, try things like a waveshaper or waveshape of an oscillator, decay of an envelope, the speed of an LFO, or parameters in effect-modules, such as reverb amount or delay time. This will create more lively patches.

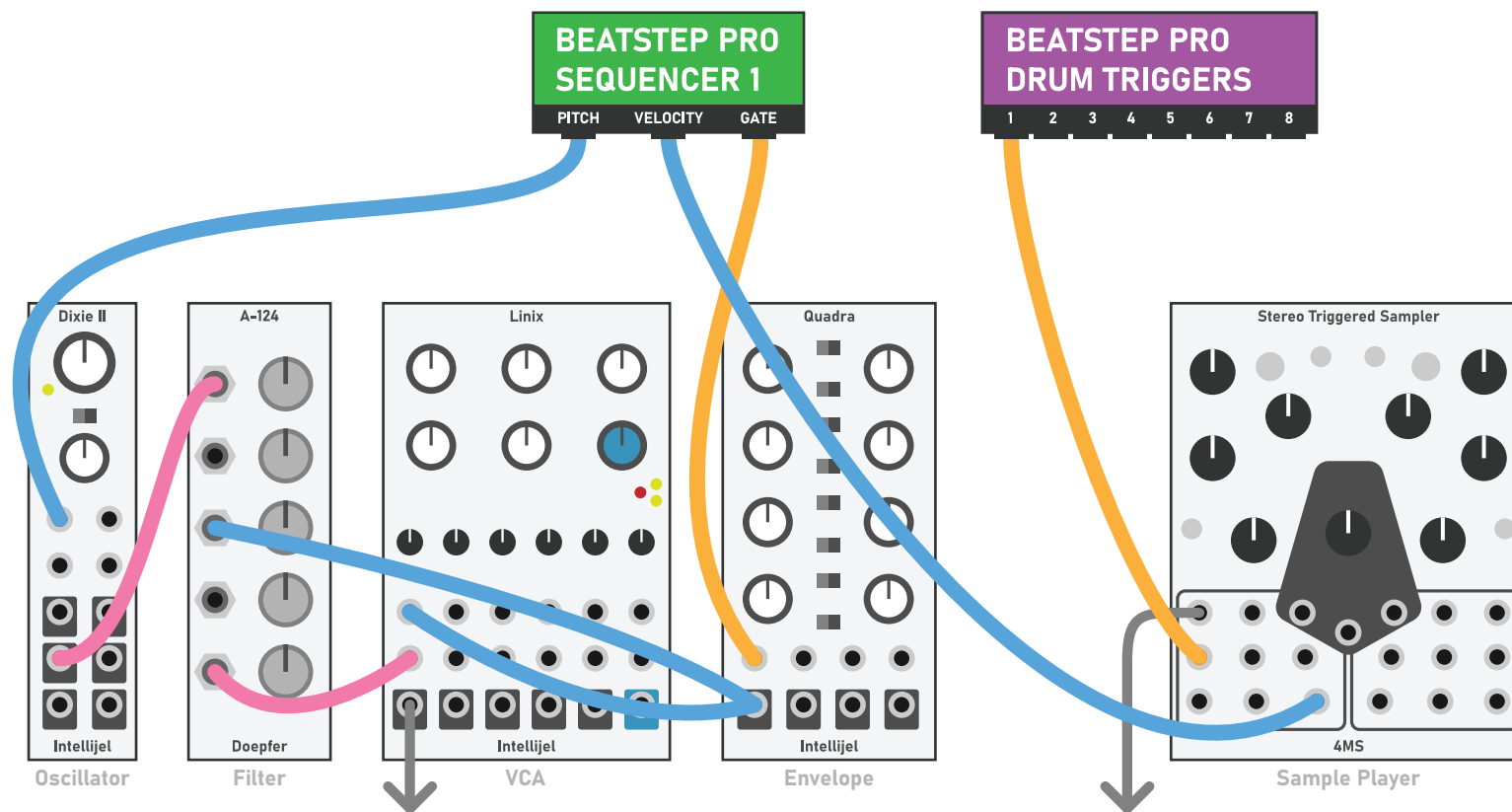
a - You can also send the velocity sequence to a VCA and control the amount of modulation passing on to something. For example, the amount of frequency modulation from a second to the main oscillator. Or the amount an LFO is modulating the filter.

Eurorack + Beatstep Pro

Creative Velocity

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(a)

a - Here's an example patch I use with a creative use for the velocity sequence. The Beatstep Pro's pitch is controlling the pitch of the Dixie II. The gate triggers an envelope on the Quadra, which is going to the Wasp Filter and Linux VCA, creating a simple voice. Then, I use the velocity sequence to the sample select input of the sampler, picking one of ten prepared single drum hits.

This way I use the velocity output of the melodic sequencer, to program entire drum sequences on a sample player. To add some complexity, you can use drum triggers from one of the 8 trigger tracks on the BSP to send a trigger pattern to the sample player.

Eurorack + Beatstep Pro

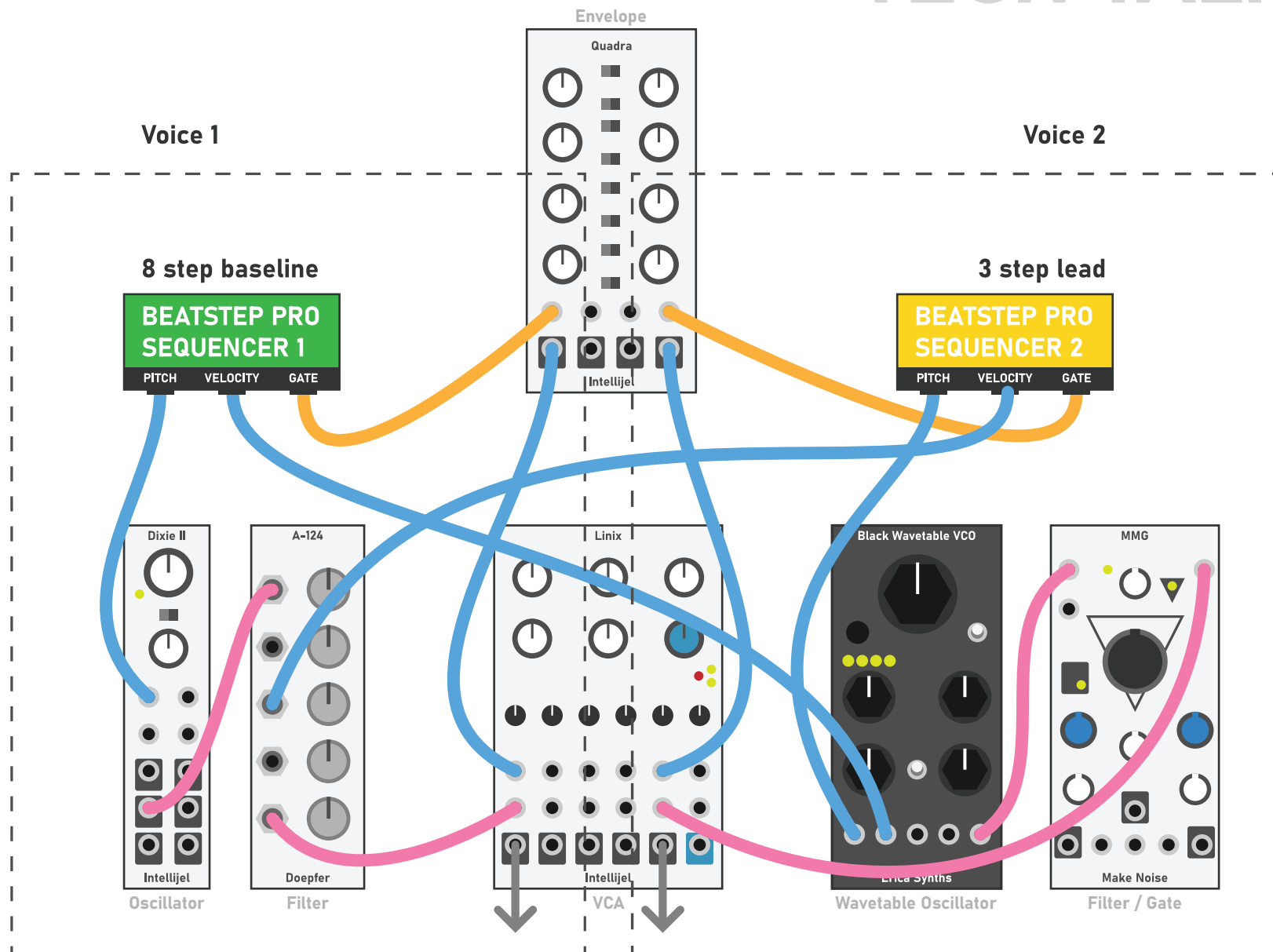
Dual Voice Patch

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate

a - In this example I use two different voices. One is an eight-step sequence controlling a baseline from the Dixie II, through the Wasp Filter and the Linux VCA. The other is a three-step sequence controlling a higher voice from the Wavetable Oscillator, through the Multi-mode Gate, and again, the Linux VCA.

What makes this patch interesting, is that the three-step velocity sequence is controlling the filter of the eight-step base, and the velocity of the eight-step lead the wave shape of the three-step melody. By controlling the amount of modulation you can change the character of each voice, related to the pattern length of the other.

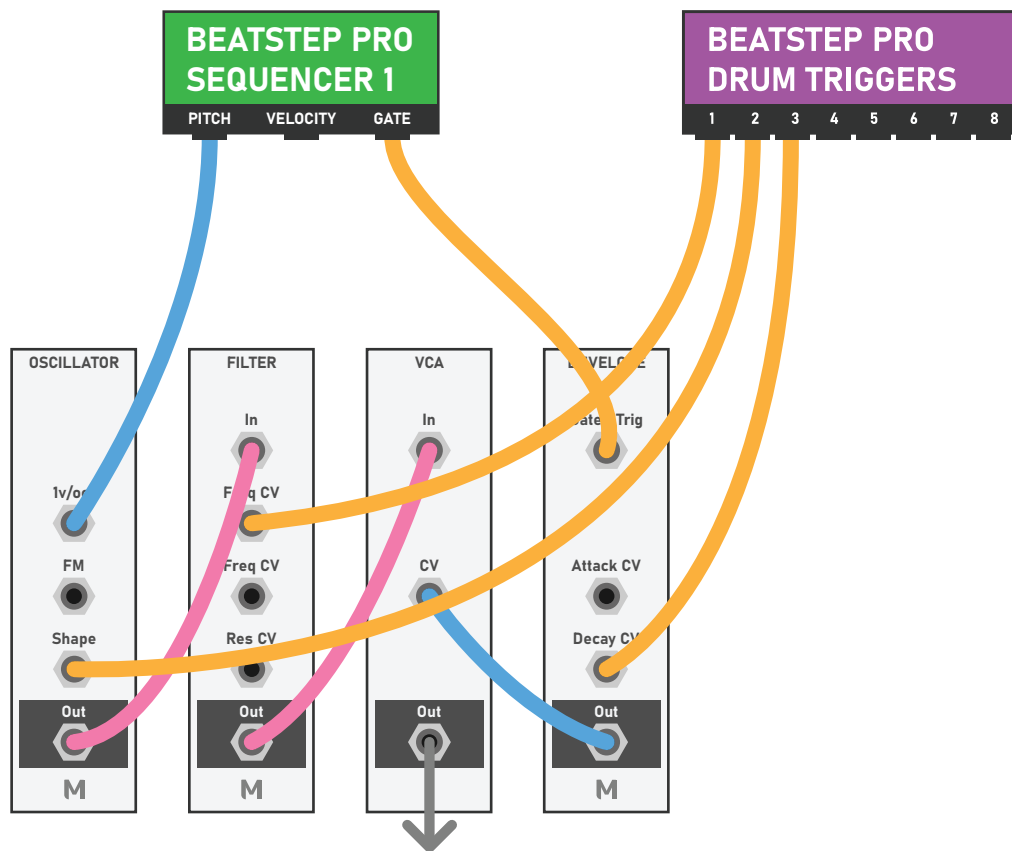


Eurorack + Beatstep Pro

Gate Patterns

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

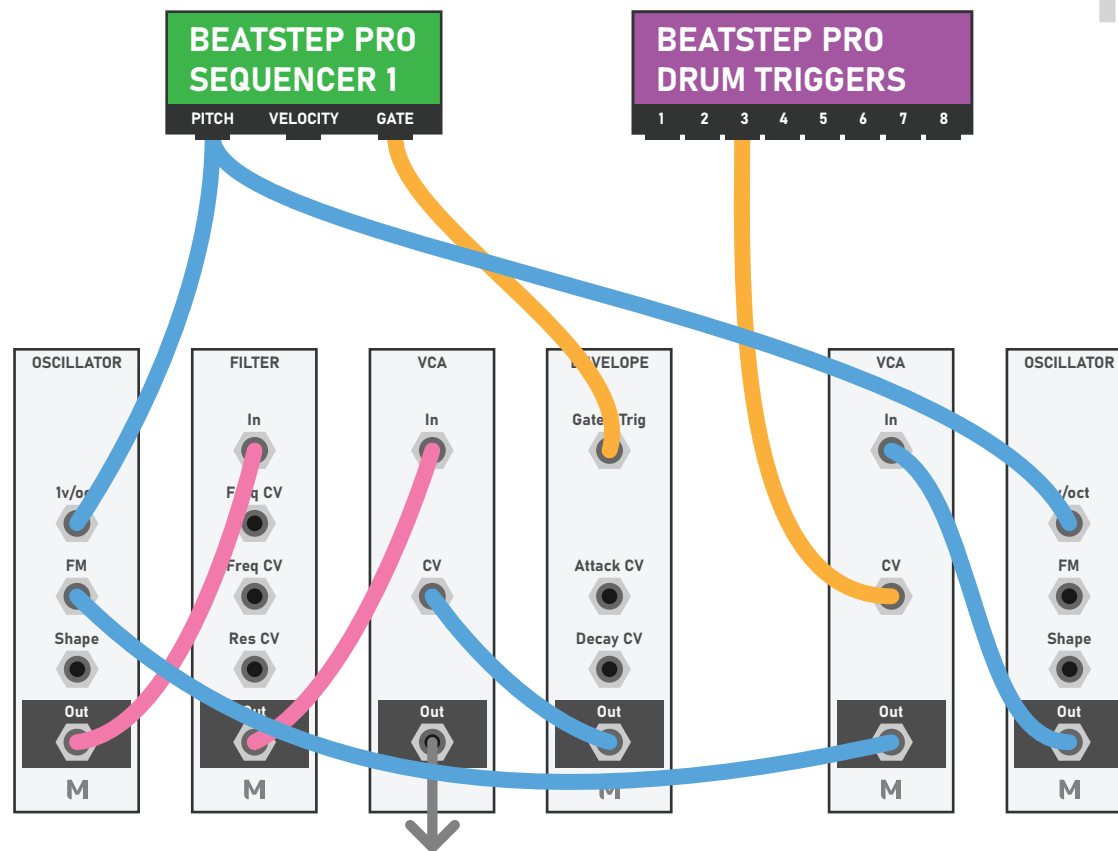
a - You can use the trigger sequences of the Beatstep Pro as one-step gates, and send them anywhere you like. Here's our simple voice again. You can use a gate pattern to place accents on a sequence by opening the filter on certain steps for example. Or think of sending the gates to the waveshape, or programming certain steps where the decay of a voice is longer.

Eurorack + Beatstep Pro

Gate Patterns

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - You can also use gates to open a VCA and create single-step bursts of modulation to something. This is especially interesting with very slow speeds. For example, create a burst of frequency modulation to the main oscillator. Or think of things like a random voltage to the delay time. Or a step of LFO to a filter. Be creative! And if you don't like the hard edges of a gate, remember you can use a gated envelope generator or slew limiter to shape them.